

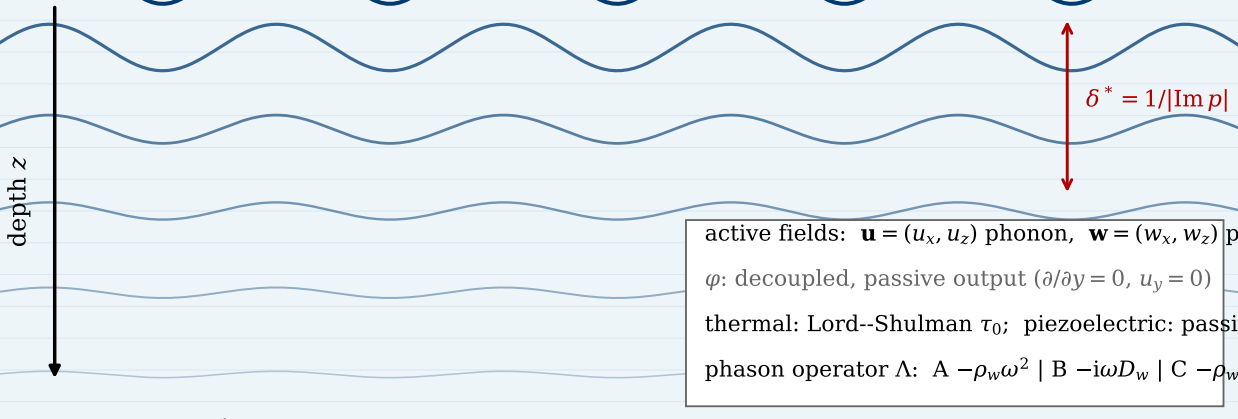
propagation $x \propto e^{i(kx - \omega t)}$



free surface: $\sigma_{zz} = \sigma_{xz} = 0$, $H_{xz} = H_{zz} = 0$ (free phason), $\theta = 0$ (isothermal)

quasicrystal half space $z > 0$ (2D hex., Laue 10)

$z = 0$



$$\delta^* = 1/|\text{Im } p|$$

active fields: $\mathbf{u} = (u_x, u_z)$ phonon, $\mathbf{w} = (w_x, w_z)$ phason, θ

φ : decoupled, passive output ($\partial/\partial y = 0$, $u_y = 0$)

thermal: Lord-Shulman τ_0 ; piezoelectric: passive in this cut

phason operator Λ : $A - \rho_w \omega^2 \mid B - i\omega D_w \mid C - \rho_w \omega^2 - i\omega D_w$

y (periodic direction, out of plane)